Session 10: EDA — Variation — Pen-and-Paper Pair Exercise

PSY 410 | Data Science for Psychology

Name: ____ **Date:** ____________________

No laptop today? Work with a partner who has one and compare answers at the end.

The data: `class_survey` (N = 34)

You'll focus on **`social_media_hrs`** — self-reported hours per day on social media.

Min	Q1	Median	Mean	Q3	Max	NAs
0.00	1.00	2.75	2.92	4.00	8.00	0

A few rows from the data:

major	year	social_media_hrs	sleep_hrs	tabs_open
Psychology	Junior	7.00	8	363
Psychology	Junior	1.00	9	2
Psychology	Other	0.33	7	7
Psychology	Senior	8.00	7	26
Neuroscience	Senior	4.50	7	6

The task (same as the slide exercise)

1. Plot the distribution of **`social_media_hrs`** — what shape is it?
2. Check for **outliers**. Are any values suspicious?
3. What does this tell you about our class?

Step 1: Sketch a histogram. Using the summary stats, sketch the shape:

```
Count
  ^
  |
  |
  |
  |
  +----|----|----|----|----|--->
       0    2    4    6    8        social_media_hrs
```

Step 2: What shape do you expect? Circle one:

Symmetric / bell-shaped • Right-skewed • Left-skewed • Bimodal

Why? (Hint: where do the median and mean sit relative to each other?) ______________________________

____

Step 3: Sketch a boxplot. Box from Q1 (*) to Q3 (*). Median at _____. Whiskers to min/max (or 1.5 × IQR, whichever is closer).

```
      |    |    |    |    |
 +----|----|----|----|----|--->
      0    2    4    6    8       social_media_hrs
```

Step 4: Outlier check. Compute the IQR and fences:

- IQR = Q3 − Q1 = _____ − _____ = _____
- Lower fence = Q1 − 1.5 × IQR = _____ Upper fence = Q3 + 1.5 × IQR = _____

Any values beyond these fences? ________ Suspicious or just honest? ____

Step 5: Write the code. Fill in the blanks:

```
# Histogram
ggplot(class_survey, aes(x = ________________)) +
  geom_____________(binwidth = 1, fill = "#2c7fb8", color = "white") +
  theme_minimal()

# Boxplot
ggplot(class_survey, aes(x = ________________)) +
  geom_____________() +
  theme_minimal()
```

Check your work. Compare your sketches, fences, and code with your partner's screen. Do your answers match what the actual plots show?